Person-Independent Wi-Fi CSI Human Activity Recognition for ESP32-Oriented Deployment

Alessandro Catenacci
IoT Systems Course Project

Abstract

This report presents a complete pipeline for human activity recognition from Wi-Fi Channel State Information (CSI), with a focus on person-independent evaluation and ESP32-oriented deployment. Raw CSI is converted into amplitude-based Doppler spectrograms, resampled to a fixed size, segmented, and classified with a compact depthwise-separable CNN. Under strict leave-one-person-out (LOPO) evaluation on 150 recordings from five persons and an additional empty-room condition, the method reaches 80.3 % segment-level accuracy and 85.3 % trace-level accuracy over four classes (`empty`, `stand`, `walk`, `jump`). The main gain over the EHUNAM [1] baseline comes from explicitly constructing Doppler signatures before classification, rather than asking a large CNN to infer motion structure almost directly from low-resolution amplitude and phase sequences. The resulting representation is visually discriminative, the model remains small enough for ESP32 constraints, and the main remaining limitation is that generalization has only been verified within a single environment.

1 Introduction and Objectives

Human activity recognition from Wi-Fi sensing aims to infer motion without requiring wearable devices or cameras. Compared with vision-based systems, CSI-based sensing can reuse existing wireless infrastructure and offers a less intrusive alternative. In this work, the goal is to classify Doppler signatures extracted from CSI under a person-independent protocol, while keeping the final model small enough for ESP32-class deployment.

The work has four practical goals: to build a reproducible preprocessing pipeline from raw CSI data to fixed-size Doppler tensors, to train and evaluate a lightweight CNN under LOPO splitting, to keep the model compatible with constrained hardware, and to provide clear diagnostic plots for interpretation of the results.

The contribution is therefore not only the classifier itself, but the full workflow linking raw CSI data, Doppler feature extraction, person-independent evaluation, and deployment-aware model design in a single reproducible setting.

1.1 What CSI Represents and Why It Is Useful

Channel State Information (CSI) is a fine-grained estimate of how a wireless signal propagates from a transmitter to a receiver. In OFDM systems such as Wi-Fi, the signal is divided across many narrow subcarriers, and CSI describes the channel response on each of them. For subcarrier k,

$$Y_k = H_k X_k + N_k, \tag{1}$$

where X_k is the transmitted symbol, Y_k is the received symbol, N_k is noise, and H_k is the complex channel coefficient.

The key idea is that H_k changes when the environment changes. Wireless signals are reflected, scattered, and attenuated by walls, furniture, and especially by the human body, so even small movements can modify the propagation paths. When CSI is observed over time, these variations reveal how motion affects the channel.

In practice, this means that different activities leave different time-frequency signatures in CSI. Empty scenes tend to be relatively stable, walking often produces more regular Doppler structure, and jumping tends to generate stronger transient patterns.

This makes CSI attractive for HAR because it enables device-free sensing, preserves privacy better than camera-based approaches, and can be obtained from commodity Wi-Fi hardware with suitable extraction support.

2 Dataset Description and Structure

2.1 Source and Scope

The data analyzed in this work comes from the EHUNAM dataset [1], focusing on the MC1 20 MHz subsets `MC1_01A` and `MC1_02A`. These subsets were selected because they better match the constraints of ESP32-class hardware than the denser 80 MHz acquisitions of the same campaign.

Compared with the 80 MHz traces, the 20 MHz subsets provide fewer CSI measurements per recording and are therefore more challenging for activity recognition. This lower temporal density also makes them a relevant setting for lightweight sensing pipelines designed for resource-constrained devices.

Technical characteristics of the campaign:

- bandwidth: 20 MHz,
- Wi-Fi standard family: IEEE 802.11n/ac,
- capture tool: Nexmon CSI Extractor (NCE),
- OFDM subcarriers per CSI frame: 64,
- temporal spacing: 100 ms (`MC1_01A`) and 50 ms (`MC1_02A`),
- approximate CSI frames per 60 s measurement: 0.6K–1.2K.

Recordings were collected in an empty basement room of 9.4 m × 13.6 m. The activity classes considered in this work are `Standing (S)`, `Walking (W)`, `Jumping (J)`, and `Empty (E)`.

Each recording contains a complex CSI matrix $\boldsymbol{H} \in \mathbb{C}^{T\times 64}$. Filenames encode the configuration, setup, scenario type (`E` vs `HAR`), person identifier, and activity label. In the implemented preprocessing, 52 data subcarriers are retained after discarding guard and pilot subcarriers.

2.2 Participant Metadata

The selected subset includes five participants (a through e). The demographic metadata provided for these participants is summarized below.

Table 1: MC1 person metadata from the EHUNAM campaign documentation.

ID	Gender	Age	Height (m)	Weight (kg)
a	Male	54	1.73	74
b	Male	54	1.71	71
c	Female	53	1.58	50
d	Male	45	1.80	85
e	Male	33	1.88	84

2.3 Class and Metadata Distribution

The processed dataset is balanced across acquisition configurations and setups, while the `empty` class is intentionally over-represented relative to each individual motion class. This asymmetry is useful in practice because it strengthens occupancy discrimination, but it also motivates the use of class-weighted optimization during training.

Table 2: Dataset composition after preprocessing

Class	Traces	Segments ($W = 50$, hop $= 25$)
empty	60	420
stand	30	210
walk	30	210
jump	30	210
Total	150	1050

3 Signal Processing and Feature Construction

3.1 From CSI to Uniform Doppler Maps

Each trace is stored as a complex CSI matrix $\boldsymbol{H} \in \mathbb{C}^{T\times 64}$, where $H_{t,k}$ denotes the CSI value at packet index t and subcarrier k. Following the implementation, only raw subcarrier indices $k = 6, \ldots, 57$ are retained, yielding a 52-subcarrier matrix used for motion-sensitive processing. In the equations below, k is re-indexed over these 52 retained subcarriers. This central crop is consistent with the common 20 MHz ESP32/802.11n practice of working with 52 usable subcarriers, although the exact raw-to-usable mapping depends on the extraction implementation.

The corresponding amplitude representation is

$$A_{t,k} = |H_{t,k}|. \tag{2}$$

Rows with zero aggregate CSI magnitude are discarded, and short-time Doppler power is computed over overlapping temporal windows. For window index n, with start position $s_n = n \cdot \text{hop}$ and length L, the amplitude samples are centered independently for each subcarrier and multiplied by a Hann window:

$$\tilde{A}_n(u,k) = \left(A_{s_n+u,k} - \bar{A}_{n,k}\right) w(u), \qquad u = 0, \ldots, L-1. \tag{3}$$

A real FFT is then applied along time for each subcarrier, and the resulting spectral energies are summed across subcarriers to obtain a one-dimensional Doppler power profile:

$$P_n(m) = \sum_{k=1}^{52} \left| \sum_{u=0}^{L-1} \tilde{A}_n(u,k)\, e^{-j2\pi m u/N_{\text{FFT}}} \right|^2, \qquad m = 0, \ldots, \frac{N_{\text{FFT}}}{2}. \tag{4}$$

The processing parameters depend on the acquisition configuration:

- `MC1_01A`: $f_s = 10\,\text{Hz}$, $L = 50$, hop $= 1$, $N_{\text{FFT}} = 128$,
- `MC1_02A`: $f_s = 20\,\text{Hz}$, $L = 100$, hop $= 2$, $N_{\text{FFT}} = 256$.

Although numerically different, these two settings are physically aligned: both correspond to an analysis window of about 5 s and yield the same nominal Doppler resolution,

$$\Delta f = \frac{f_s}{N_{\text{FFT}}} \approx 0.078\,\text{Hz}.$$

This preserves comparable motion-frequency content across the two recording regimes.

After spectral estimation, the DC component is suppressed by setting $P_n(0) = 0$, and only frequencies up to 5 Hz are retained. Because both configurations share the same Doppler resolution, this crop yields 65 bins in each case, including the DC bin. The resulting time–frequency map is resampled with first-order interpolation to a common tensor

$$\boldsymbol{D} \in \mathbb{R}^{200\times 65},$$

where 200 is the chosen uniform time dimension and 65 is the shared low-Doppler frequency dimension. This removes differences caused by variable trace duration and configuration-dependent sampling density, allowing all traces to be processed by the same classifier.

3.2 Training-Time Normalization and Segmentation

The uniform Doppler tensor $\boldsymbol{D}$ is not fed directly to the classifier. Before training, each map is log-compressed and standardized:

$$\boldsymbol{D}' = \log(1 + \boldsymbol{D}), \qquad \hat{\boldsymbol{D}} = \frac{\boldsymbol{D}' - \mu_{\boldsymbol{D}'}}{\sigma_{\boldsymbol{D}'} + \epsilon}, \tag{5}$$

where $\mu_{\boldsymbol{D}'}$ and $\sigma_{\boldsymbol{D}'}$ are the scalar mean and standard deviation of the full trace-level map. The logarithm reduces dynamic range, while per-trace standardization limits sensitivity to absolute energy differences.

Each normalized map is then split into overlapping temporal segments of 50 bins with 25-bin hop, producing 7 segments per trace. This both augments the training set and lets the CNN focus on shorter regions where motion patterns are typically easier to discriminate. Examples are shown in Figure 2.

4 Model Architecture and Training Protocol

4.1 CNN Design

The classifier is a depthwise-separable 2D CNN with 4,516 parameters. This design was chosen to retain local time-frequency modeling while keeping the model compact and computationally lightweight. It is also appropriate for the MC1 20 MHz subsets, where each trace contains only about 0.6K–1.2K CSI frames, making larger CNNs more prone to overfitting. As shown in Figure 1, the resulting architecture remains compatible with ESP32 constraints, requiring about 4.4 KB for int8 model storage, about 203.1 KB peak activations, and an estimated int8 latency of 56 ms.

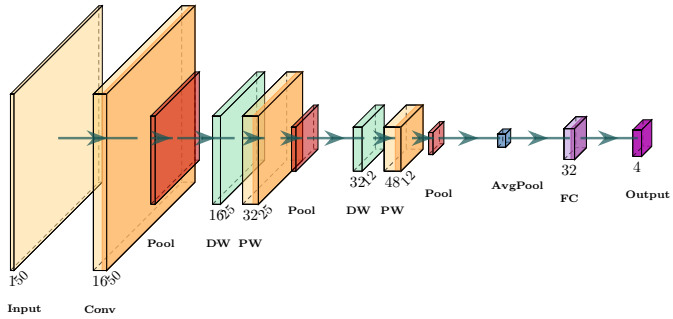

Figure 1: Compact depthwise-separable CNN topology used in all LOPO folds.

4.2 Optimization

Training uses AdamW (learning rate 10^{-3}, weight decay 10^{-3}), cosine annealing, gradient clipping (norm 1.0), and early stopping (patience 15), with a maximum of 60 epochs.

Class imbalance is handled by inverse-frequency weighting:

$$w_c = \frac{\frac{1}{n_c}}{\sum_{j=1}^{C} \frac{1}{n_j}} \cdot C, \tag{6}$$

and weighted cross-entropy:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^{B} w_{y_i} \log p_{i,y_i}. \tag{7}$$

Data augmentation includes random circular shifts along time, additive Gaussian noise, and random frequency masking to improve robustness to small temporal misalignments and moderate spectral variability.

4.3 Person-Independent Evaluation

Evaluation follows a leave-one-person-out (LOPO) protocol, splitting the dataset into five folds, one for each person (a to e). In each fold, the model is trained on four persons and tested on the remaining unseen one. This is stricter than a random train/test split because it measures whether the classifier has learned activity-dependent motion patterns rather than person-specific habits.

The empty-room class requires a separate rule because it has no person identity. These traces are grouped by (config, setup, repetition) and assigned fold-wise so that recordings from the same physical acquisition session remain together and never appear in both training and test. The leakage audit found no leakage in any fold, so the reported metrics can be interpreted as genuine person-independent estimates rather than optimistic results caused by within-person or within-session reuse.

For auxiliary comparison with the EHUNAM paper, the same Doppler pipeline was also evaluated under the reference-style random 65/17.5/17.5 split.

5 Qualitative Characterization of Doppler Signatures

Before analyzing quantitative performance, it is useful to inspect whether the Doppler representation exhibits class-specific structure that a lightweight CNN could plausibly exploit. Figure 2 presents representative spectrograms for the four target classes after preprocessing and resampling.

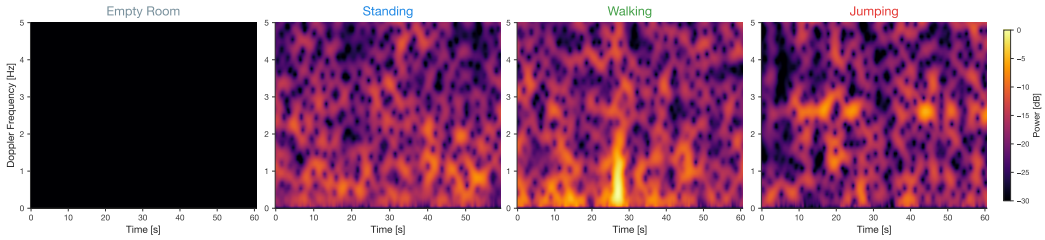

Figure 2: Representative Doppler spectrograms for the four target classes after preprocessing. Empty and standing traces remain concentrated near the low-Doppler region, whereas walking and jumping produce broader temporal and spectral excursions.

The figure supports the central modeling assumption of this work: once the CSI is converted into Doppler energy maps, the velocity structure becomes explicit enough that the classes are already partially distinguishable by eye. This is a major difference from pipelines that feed low-resolution amplitude and phase streams almost directly to the network. At the same time, the qualitative overlap between walking and jumping already suggests that the most difficult errors will arise among the dynamic classes rather than between occupied and empty conditions.

6 Results

6.1 Fold-Wise LOPO Performance

Figure 3 summarizes LOPO performance from two complementary views: per-fold accuracy at segment and trace level, and the corresponding class-wise accuracy across held-out persons. Trace-level predictions are obtained by aggregating the logits of all segments belonging to the same trace. This usually improves person-wise performance and clearly improves the aggregate metric, indicating that the model often captures useful local evidence even though single windows may remain ambiguous when classified in isolation.

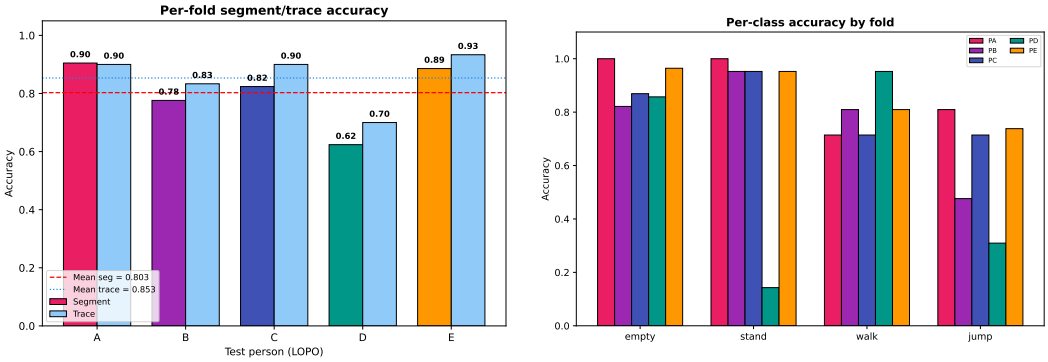

Figure 3: Compact view of LOPO accuracy trends.

Across folds, accuracy increases from 80.3 % at segment level to 85.3 % after trace-level aggregation, indicating that many errors reflect local window instability rather than full-trace failures. The held-out-person spread remains noticeable, with segment accuracy ranging from 62.4 % for person D to 90.5 % for person A, suggesting that some subjects exhibit motion patterns less aligned with those learned from the others.

6.2 Class-Level Metrics

Table 3 and Table 4 report precision, recall, and F1 score at segment and trace level, respectively. The main pattern is a clear class asymmetry: `empty` is nearly saturated, `stand` remains relatively stable, and the dominant ambiguity is concentrated between the two dynamic classes. In other words, the main residual difficulty is not separating static from non-static conditions, but distinguishing different forms of motion under person-independent evaluation.

Table 3: Classification report summary (segment-level).

Class	Precision	Recall	F1-score
empty	0.96	0.90	0.93
stand	0.86	0.80	0.83
walk	0.64	0.80	0.71
jump	0.65	0.61	0.63

Table 4: Classification report summary (trace-level mean-logit voting).

Class	Precision	Recall	F1-score
empty	1.00	0.95	0.97
stand	0.96	0.83	0.89
walk	0.66	0.83	0.74
jump	0.72	0.70	0.71

At segment level, `empty` already achieves very strong precision and recall, showing that the absence of motion is represented distinctly. `Stand` also remains well behaved, with comparatively balanced metrics. By contrast, the dynamic classes are less symmetric: `walk` is recall-heavy and precision-limited, whereas `jump` is more recall-limited, especially at segment level. This suggests that part of the stronger motion variability is absorbed into the broader `walk` category, causing some `jump` instances to be missed when only short windows are considered.

Trace-level aggregation reduces this instability. The clearest effect is observed for `jump`, whose F1 score rises from 0.63 to 0.71, indicating that burst-like motion becomes more separable once evidence is accumulated over the full trace. `Stand` also benefits from aggregation, while `empty` remains near-perfect. Overall, sequence-level voting improves the aggregate accuracy and yields a more balanced class-wise profile, even though the gain is not strictly monotonic in every held-out fold.

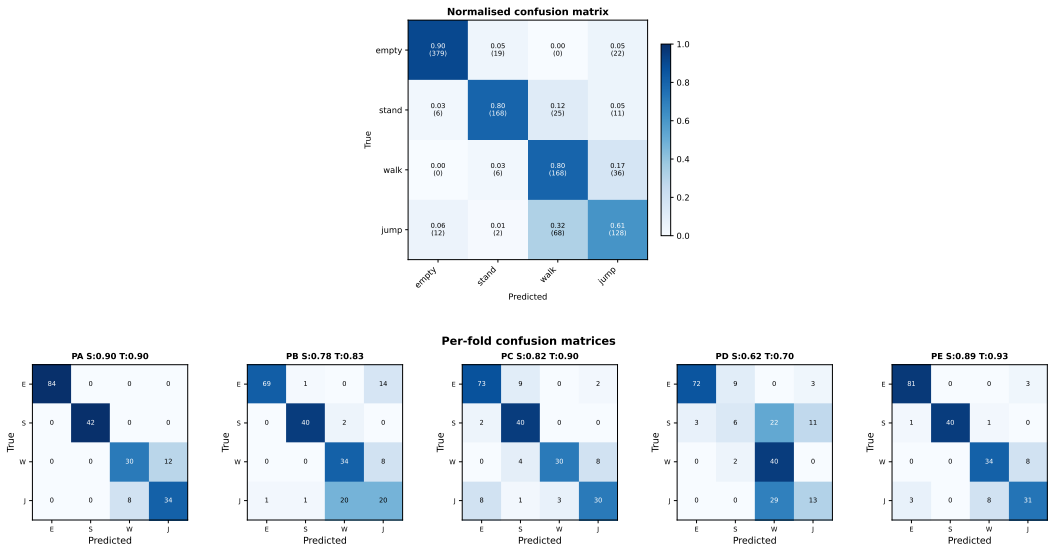

Figure 4: Confusion analysis under LOPO evaluation. Most off-diagonal mass is concentrated between walking and jumping, while the per-fold view shows that this ambiguity is markedly stronger for some held-out persons than for others.

The confusion matrices in Figure 4 are consistent with the metric tables. The `empty` class is separated cleanly, while most of the off-diagonal mass is concentrated between `walk` and `jump`. The per-fold view further shows that dynamic-class ambiguity varies across held-out persons, indicating that inter-person motion variability remains a relevant factor under LOPO evaluation.

7 Comparison with the Reference EHUNAM Pipeline

All comparisons in this section were performed on the same EHUNAM `MC1_01A` and `MC1_02A` 20 MHz subsets used in the present study.

7.1 Reference Pipeline Summary

For the MC1 20 MHz subsets, the EHUNAM reference workflow follows a phase-centric deep-learning pipeline:

1. **TSFR preprocessing**: linear-regression phase detrending, Savitzky–Golay temporal smoothing, threshold-based phase discontinuity correction, and null-subcarrier removal.
2. **CNN model**: a TensorFlow/Keras architecture with three convolutional stages (32 filters 5×5, 64 filters 3×3, 128 filters 3×3), pooling/dropout regularization, and dense layers of 256 and 128 units, using Mish activations and Softmax output.
3. **Training split**: 65% train, 17.5% validation, 17.5% test, Adam optimizer (10^{-3}), batch size 256 for 20 MHz data.

The reported input shape is $(S, 25, 2)$, where S denotes the number of useful subcarriers (56 in those subsets) and the two channels correspond to amplitude and phase. Overall, this design feeds calibrated amplitude–phase CSI to a relatively large CNN, but does not construct an explicit motion-domain representation such as a Doppler map. Temporal discrimination is therefore learned mostly by the network itself.

7.2 Published MC1 20 MHz Results

Table 5: Reported accuracies for the MC1 subsets from the EHUNAM publication.

Dataset subset	Temporal spacing	Reported accuracy
`MC1_01A`	100 ms	68.4%
`MC1_02A`	50 ms	59.7%

These values are lower than those reported for the denser 80 MHz subsets in the same campaign, which is consistent with the fact that the 20 MHz traces contain fewer CSI packets over the same 60 s interval. The classifier therefore receives a temporally poorer motion signal, making fine activity discrimination more difficult.

7.3 Side-by-Side Comparison with the Present Pipeline

The scope of this comparison is twofold: first, to benchmark the present pipeline against the EHUNAM reference on the same subsets under a protocol-matched random split; second, to report the main performance of the present method under the stricter LOPO protocol used throughout this report.

Table 6: Methodological comparison between the reference 20 MHz pipeline and the present approach.

Aspect	**Reference paper pipeline**	**Present pipeline**
Input	Amplitude + phase	Amplitude only
Preprocessing	TSFR phase calibration and smoothing, without explicit Doppler representation	Doppler STFT on amplitude, DC notch, $f \leq 5$ Hz, uniform resampling to a fixed 200×65 grid
Model size	~ 820 k parameters	4,516 parameters
Evaluation	Random 65/17.5/17.5 split	LOPO by person
20 MHz **result**	`MC1_01A`: 68.4% `MC1_02A`: 59.7% Average: 64.0%	Reference-matched random: 78.9% LOPO segment: 80.3% LOPO trace: 85.3%

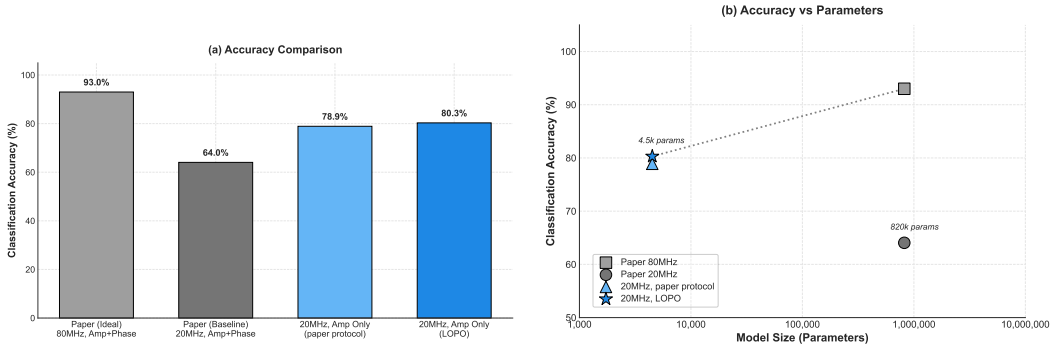

Figure 5: Comparison between the reference 20 MHz pipeline and the present amplitude-only approach. The left panel highlights protocol-level accuracy differences, while the right panel situates the proposed model on the accuracy-efficiency trade-off frontier.

The 78.9% reference-style result in Table 6 and Figure 5 was obtained by rerunning the present Doppler pipeline under the same random 65/17.5/17.5 split used in the reference paper and averaging over 10 random seeds. Its purpose is simply to provide a protocol-matched comparison with EHUNAM. The 80.3% value, by contrast, comes from the updated deterministic LOPO experiment, which remains the main evaluation setting of this report.

The important point is not the numerical ordering between the random-split and LOPO scores, but the high variability of the random-split estimate on these subsets. Some seeds performed better than the LOPO score, as one would expect from an easier protocol, while others underperformed substantially. This is plausible given the small size of the selected 20 MHz dataset: with only 150 traces, random splitting is highly sensitive to which person styles, trace repetitions, and acquisition conditions fall into train, validation, and test. In other words, the protocol-matched random split remains useful for comparison with the paper, but on this limited subset it is a noisier estimate than structured LOPO evaluation.

7.4 Why the EHUNAM Baseline Underperforms at 20 MHz

The main issue is representational. In the reference pipeline, the CNN is asked to infer motion structure almost directly from amplitude and phase sequences after limited calibration and smoothing. That is already demanding in richer datasets, and becomes especially problematic at 20 MHz, where each trace contains fewer CSI samples than the denser subsets in the same campaign. The result is a weaker and more irregular motion signal from which the classifier must extract activity structure.

In the present pipeline, this burden is reduced before learning. Doppler processing makes the velocity signature explicit, so the network no longer needs to discover the relevant time–frequency organization from nearly raw measurements. This is also why the classes become visually distinguishable in Figure 2: the preprocessing already concentrates the motion evidence into a task-aligned representation.

Two further differences likely contribute. First, each spectrogram is resampled to a fixed 200×65 grid, which makes the input geometry uniform across configuration differences. Second, classification is performed on overlapping temporal segments rather than on the entire trace at once, increasing the number of training samples while localizing the most informative motion regions.

Model scale is the other likely factor. The reference CNN is large and reused across datasets, whereas the MC1 20 MHz subsets are relatively small. In that regime, a high-capacity network is more exposed to overfitting idiosyncratic patterns. The present depthwise-separable CNN is deliberately compact, making it better matched to the available data while also satisfying the embedded deployment constraint.

Accuracy delta against 20 MHz references. Using segment-level LOPO accuracy as the primary score, the present method improves by 16.3 points over the 64.0% paper baseline shown in Figure 5. The comparison plot also shows that the method remains strong under the stricter person-independent protocol rather than only under a random split.

8 Limitations and Multi-Environment Generalization

The current pipeline has clear limitations:

1. the data comes from a single environment, limiting domain diversity;
2. only five persons are available for person-independent evaluation;
3. the class set is limited to `empty`, `stand`, `walk`, and `jump`;
4. the amplitude-only representation may discard phase information useful for fine-grained motion discrimination.

The concentration of errors in a single held-out person and in the `jump` class confirms that robustness is not homogeneous across all motion styles. More broadly, the present study demonstrates person-independent generalization only within one room and one acquisition campaign.

In principle, the Doppler-centered representation should transfer across environments better than raw CSI because it suppresses part of the static channel geometry and emphasizes motion-induced frequency shifts, which are more directly related to human velocity than to a specific room response. The fixed-grid resampling also removes part of the device- and acquisition-dependent variability in input size. These choices should reduce the domain gap relative to an almost raw amplitude-phase pipeline, but they cannot eliminate it: room layout, transceiver placement, multipath richness, wall materials, and person-to-sensor geometry still affect the strength and spread of the Doppler energy.

For a true multi-environment evaluation, the appropriate next protocol would be leave-one-environment-out or train-on-several-environments/test-on-unseen-environment using a dataset with multiple rooms, placements, and sessions. Under that setting, the present pipeline would be expected to retain its main advantage, namely a more invariant motion representation, but likely with a measurable drop relative to the single-room LOPO result because environment-specific Doppler envelopes would still remain. In other words, this work supports a plausible path toward better environment generalization, but it does not yet prove it empirically.

9 Deployment Feasibility

Embedded feasibility was assessed through an analytical memory-budget estimate, obtained by summing (number of elements) $\times$ (bytes per element) over the main pipeline buffers and intermediate representations. For example, a 200×65 floating-point map requires 52,000 bytes ($\approx$50.8 KiB), while the segmented CNN input requires 91,000 bytes ($\approx$88.9 KiB); applying the same calculation to the full preprocessing state gives a total of about 618.6 KiB.

Together with the compact CNN, this is consistent with the memory budget of the target ESP32-S3 platform and supports deployment feasibility from a resource perspective. Hardware runtime validation is left outside the scope of this work.

10 Conclusion

The implemented CSI-to-Doppler pipeline shows that reliable person-independent HAR can be obtained under a strict LOPO protocol with a model small enough for ESP32-class deployment. The main result is not only the achieved accuracy, but the observation that explicit Doppler construction, fixed-grid resampling, overlap-based segmentation, and a compact CNN form a much better match for sparse 20 MHz CSI than a large network trained close to the raw signal. The dominant residual errors remain concentrated in dynamic-class separation and in a small subset of difficult held-out persons. The next substantive step is therefore not scaling the CNN, but testing whether the same representation remains stable across rooms, placements, and sessions in a true multi-environment benchmark.

References

[1] E. de Armas, G. Diaz, I. Sobron et al., "EHUNAM, a WiFi CSI-based dataset for human and machine sensing," *Scientific Data*, vol. 12, no. 1, p. 1950, 2025. doi: 10.1038/s41597-025-06238-4.